

STORY AGENT · PILLAR

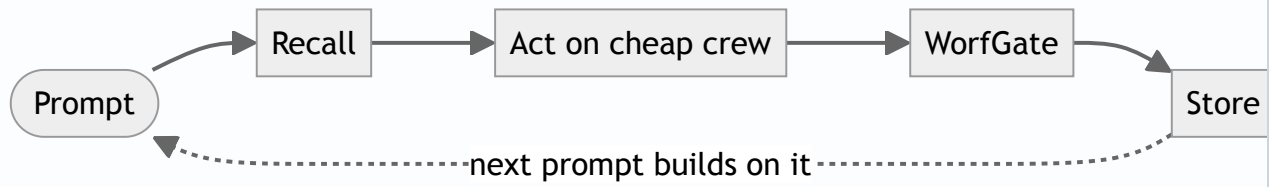
Memory & Learning

How this pillar works and why it matters.

Team assembled by Riker: Jean-Luc Picard, Data, Beverly Crusher.

[↑ Story Agent overview \(HTML\)](#) · [↑ overview \(PDF\)](#).

At a glance



Orchestrated Learning at Scale

- Cost-Optimized Execution: Quark routing dynamically selects the cheapest adequate model per task, with Anthropic reserved for tier-4 complexity, ensuring deliberation costs remain at ~\$0.002-0.003 per task.
- Continuous Improvement: RAG (pgvector) enables recall→act→store cycles, compounding knowledge across executions while maintaining portability and auditability.
- Governed Autonomy: WorfGate enforces credential security and auto-remediation (green/yellow/red), while async-status provides live progress tracking and timeout guardrails.

Why it matters: Delivers enterprise-grade autonomous coding at startup economics, with transparent cost control and incremental learning.

Memory Architecture: Recall, Act, Store

- pgvector RAG enforces a strict recall→act→store loop: every deliberation reads relevant context before execution and writes structured outcomes back, so the system compounds knowledge rather than restarting cold
- 41 skill theories are discrete, versioned knowledge units — each scoped to a bounded domain, preventing cross-contamination between skill contexts and enabling targeted updates without full-corpus retraining
- Portable memory layer is self-hosted on AWS Fargate with no vendor lock-in: the pgvector store travels with the deployment, and the MCP open standard means accumulated context is accessible across Claude Code, Cursor, and Copilot without re-ingestion

Why it matters: Every deliberation at ~\$0.002–\$0.003 is cheaper and more accurate than the last because the system retains structured outcomes — cost decreases as the knowledge base matures.

System Health and Learning

- Monitoring of agent performance and system latency to identify potential bottlenecks and areas for optimization
- Implementation of health checks and automated remediation strategies to ensure high availability and reliability
- Integration of logging and metrics to inform data-driven decisions and improve overall system well-being

Why it matters: By prioritizing system health and learning, clients can expect increased uptime, faster issue resolution, and improved overall platform performance.

END OF PILLAR

Memory & Learning — recap

- **Jean-Luc Picard:** Orchestrated Learning at Scale
- **Data:** Memory Architecture: Recall, Act, Store
- **Beverly Crusher:** System Health and Learning

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